TSTrans: Temporal-Sequence-Driven Transformer for Single Object Tracking in Satellite Videos

Yitong Li, Chen Wu, *Member, IEEE*, and Jia Shao

***Abstract*—Satellite videos capture a broad field of view of city-scale scenes, while moving objects of interest can only claim to be tiny, occupying only several pixels. Besides, noises from target-similar surroundings and illumination change exist everywhere in the complex satellite imaging backgrounds, where objects are easily occluded by man-made buildings or clouds. Therefore, considering the limited information capacity of spatial appearance and semantic features, satellite video object tracking (SVOT) algorithms should make the most of temporal-sequence information to cope with the above challenges involving object detail deficiency, object occlusion, and environmental noises. In this article, we focus on the critical characteristic "temporal sequence" and fully exploit its implied target changing tendency and motion continuity to boost tracker performance. The proposed temporal-sequence-driven transformer tracker (TSTrans) utilizes the Transformer framework to take advantage of its long-term relation modeling ability. To obtain robust object understanding, TSTrans introduces multiple temporal templates to extract target variation patterns, thereby achieving dense target-aware attention between searching patches and long-term template sequences. For global refinement of tracking results, TSTrans resists occlusion and model attention distraction challenges through trajectory prediction based on historical frame sequence. Comprehensive experiments have been conducted on three large-scale satellite video datasets, including SatSOT, SV248S, and VISO. Contrasted with 12 advanced trackers, our TSTrans demonstrates state-of-the-art (SOTA) tracking accuracy, with precision and success scores of 71.00%/53.20%, 86.36%/40.84% and 65.38%/28.05% on SatSOT, SV248S, and VISO, respectively, running at around 50 frames per second (FPS). Code is at https://github.com/yitongli123/TSTrans**

***Index Terms*—Satellite videos, single object tracking (SOT), temporal sequence, vision transformer (ViT).**

I. Introduction

SATELLITE video object tracking (SVOT) [1], [2], [3], [4], [5], [6] was born to be a significant vision subject, considering its widely practical applications [7]: e.g., given a satellite-captured imagery sequence, it is of great importance to focus on the state changing of vehicles, airplanes, trains and ships for intelligent traffic management [6], civil and military security [3], smart city [2], ocean monitoring [5], and so on.

However, without humans' visual perception and life experience, it is difficult for AI to find, identify, and memorize such densely distributed and tiny objects within an extensive and complex scope, covering an area of even 3768×1987 m on the Earth, for which the reasons can be summarized as the following four points.

1) *Object Detail Deficiency (Small Target Size and Low Image Resolution):* Because poor image quality causes blurry boundaries and texture, objects usually have weak appearance features by occupying only a few pixels.
2) *Object Changes (Occlusion, Deformation, and Rotation):* Objects in the satellite video are likely to be fully or partly occluded by things such as bridges, buildings, shadows, and clouds. Target's appearance may also change, such as a deformed or rotated object due to satellite imaging. All of these interferences demand appropriate target knowledge updating.
3) *Scene Distractions:* Target-similar objects belonging to the same category, confusing background components with color and texture alike. Due to the target's attribute of gathering, like crowded vehicles at the crossroads, and scene complexity, trackers tend to mistake the above surrounding distractions for the target.
4) *Environmental Noises (Illumination Variation and Background Clutter):* Besides much saliency interference from a cluttered environment, illumination and other weather factors will also affect the imagery brightness, thus making a negative influence on target discrimination.

In the past several years, more and more large-scale satellite video datasets annotated for object tracking [8], [9], [10] have been released, providing a brilliant opportunity and convenience for training convolutional neural network (CNN)-based trackers. Some advanced Siamese network trackers [1], [6], [11], [12], [13] and correlation filter (CF)-based trackers [14], [15], [16], [17], [18] employing CNNs have shown nice performance in model accuracy and inference speed. However, the correlation operators in the above CNN-based trackers are still not enough to recognize some nonlinear changes like deformation, rotation, and occlusion.

In recent years, the attention mechanism in Transformer [19] has been introduced into single object tracking (SOT) for augmenting and fusing image features. In TransT [20], multihead self-attention is separately applied to template and

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search vectors, integrating global context, while multihead cross-attention is naturally employed to calculate semantic association. For higher running speed, HiFT [21] makes use of LeViT [22] and a U-Net-like structure to bridge a lightweight hierarchical Transformer with a tracking pipeline, where dual patch embeddings are concatenated to accept solely self-attention for abundant interactions. In accordance with the encoder–decoder framework, STARK [23] utilizes an encoder to strengthen feature dependency among concatenated vectors. Its decoder receives one learnable target query, which attends to the encoder output to estimate the single object, inspired by the end-to-end object detector DETR [24]. While these CNN-Transformer algorithms [20], [21], [23], [25] can enhance the model's target awareness to a certain degree, their two-stream Siamese feature extraction is actually not target-adaptive, and their two-stage CNN-Transformer structure is evidently inefficient. Accordingly, fully Transformer approaches [26], [27], [28], [29], [30], [32] were proposed, noticing that vision transformer (ViT) [31] is able to realize adaptive inference and joint feature processing with a single encoder. For the first time, OSTrack [26] merges feature extraction and relation modeling into one stage and discards Siamese-like redundant flows through plain linear transformation and stacked self-attention in ViT [31]. Following OSTrack's encoder architecture [26], SeqTrack [27] constructs a new sequence-to-sequence tracking paradigm by replacing the common head network with an autoregressive decoder outputting a sequence of bounding box variables. To adapt to two modalities for RGB-thermal tracking, STTrack [28] designs a dual-stream backbone with independent Transformer-encoders for RGB and TIR features, where some trainable parameters are added for capability transfer, after which the two features are fused for cross-modal interaction.

However, most of the existing models, including fully CNN trackers, CNN-Transformer trackers, and fully Transformer trackers, are still appearance-oriented and constrained by static target matching within the current frame. Such appearance-driven algorithms primarily demand abundant sources of target appearance features, thus being unsuitable for satellite video objects with poor imaging details and low vision saliency. Also, these methods commit to better localizing ability solely from exploring more spatial or contextual information, thus ignoring another essential characteristic of SVOT: temporal sequence. Other than image pairs for the degraded subtask of object searching, satellite videos as frame streams provide important information about the target's continuous change and motion. These temporal cues can significantly promote the model's target tracking, yet are not comprehensively employed by the current methods. Specifically, some temporal-information-assisted algorithms only see short-range changes, not the target's varying tendency implied in long-term historical sequence, causing failure to improve object understanding. For instance, STRCF [15] introduces a temporal regularization that is constructed just between consecutive frames into discriminative CFs (DCFs). ProContEXT [29] takes multiscale versions of the dynamic template (DT) into consideration, increasing attention interference from redundant background. Although with a sequence-to-sequence decoder, SeqTrack's attention scope [27] is still limited to the given template. Most recently, there are also some frame-by-frame temporal methods: ODTrack [30] propagates a temporal token through continuous frames to accumulate video-level target attention, neglecting the low information capacity of a single token and the potential catastrophic forgetting problem. STTrack [28] updates an online template to complement temporal knowledge, while failing to capture complete temporal perception through single target change. LMTrack [32] temporally collects and updates high-importance reference tokens, removing near-target background tokens with spatial contextual information in referenced frames that benefit target recognition.

Therefore, in this article, we propose temporal-sequence-driven transformer tracker (TSTrans) investigating both target's changing tendency and its motion continuity from temporal sequence of video frames, which drives our model to exploit comprehensive temporal cues, as shown in Fig. 1. Aiming to make full advantage of satellite video's temporal sequence, we choose Transformer-based framework [19] considering its long-term relation modeling capability. Meanwhile, in order to maintain high effectiveness and efficiency of spatial feature extraction, we select the fully Transformer model OSTrack [26] as a pipeline. Since the previous template's representative capacity is limited, dynamic variation and multiple states should be considered by template sequence modeling to obtain robust target perception. Thus, we upgrade the relation encoding session to a dense temporal inter-token attention, where multitemporal target-aware similarities between the current frame and multiple temporal templates are integrated. Notably, temporal templates are reliably selected and sequentially iterated from historical frames, thus forming a long-term template sequence together with the standard template, which covers key information ranging from the initial frame to the current frame. On the other hand, by virtue of the target's motion continuity, the historical frame sequence is used for target trajectory prediction and tracking outlier avoidance. A motional correction module is designed to resist scene distractors, where we simulate the target's motion with reference to historical movements and correct tracking results for timely attention restoration when detecting trajectory departure.

We have carried out extensive experiments on three challenging satellite video datasets, SatSOT [8], SV248S [9], and VISO [10]. Experimental results display our method's best performance compared with 12 state-of-the-art (SOTA) trackers, running at around 50 frames per second (FPS). TSTrans's precision and success scores reach 71.00%/53.20%, 86.36%/40.84%, and 65.38%/28.05%, exceeding the second ranking trackers by 3.68%/4.76%, 6.86%/3.90%, and 1.96%/0.73%, on SatSOT [8], SV248S [9], and VISO [10], respectively. Our main contributions can be summarized as follows.

1) To embrace the target's dynamic variation, we exploit the underlying target changing tendency from a long-term template sequence with multiple temporal templates to alleviate the tracker's knowledge lag. Dense target-aware attentions are extracted between the search region (SR) and multitemporal target states, producing

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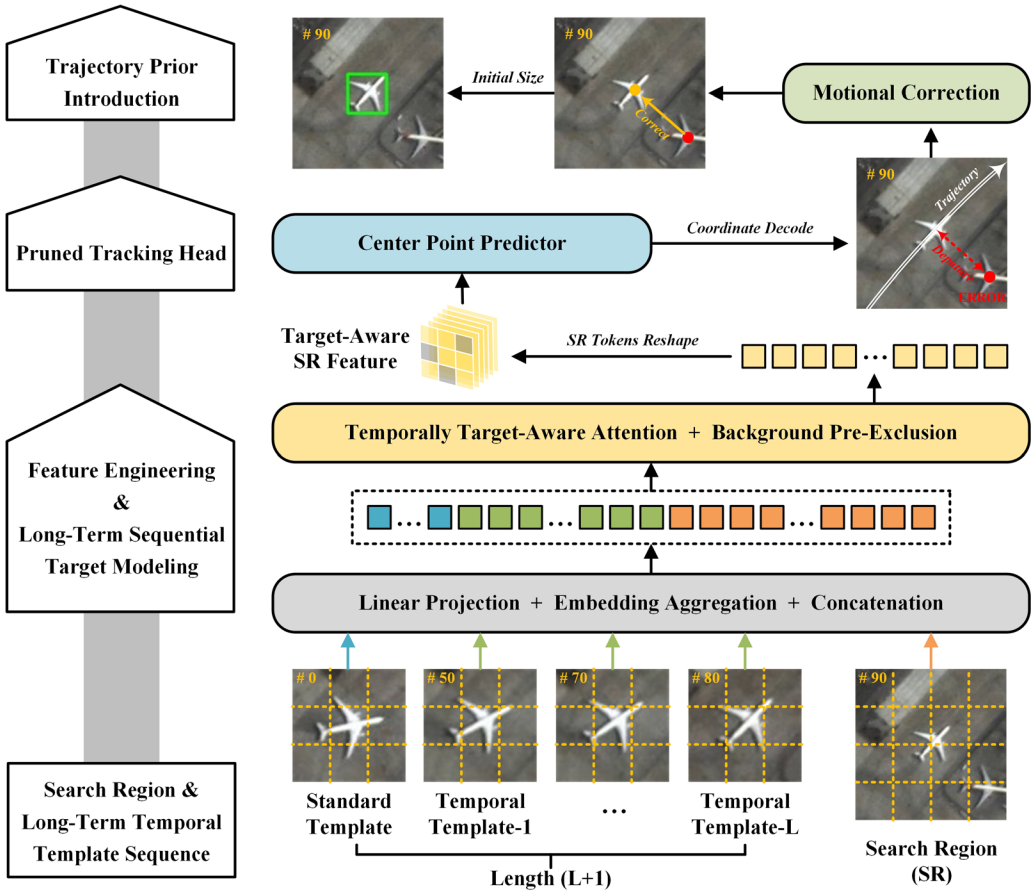


Fig. 1. Overview of the proposed TSTrans. Conforming to a one-stream one-stage tracking pipeline, TSTrans distills the target's spatial, contextual, and temporal information from a long-term temporal template sequence, which is constructed with a standard template in the initial frame and L temporal templates that are representatively updated from the historical video stream. After effective feature extraction and enhancement carried out in feature engineering, dense target-aware attentions are comprehensively modeled between SR tokens and template sequence tokens. Based on the produced target-aware SR feature, the target center is intently localized through a center point predictor and globally refined by a trajectory-based motional correction module.

a more complete target modeling. Through confidence-based sliding selection, we guarantee temporal templates' positive influence and representativeness, thus greatly improving model robustness.

2) To integrate the target's continuous motion from a historical frame sequence, we apply a motional correction module carrying trajectory simulation and global tracking error rectification, so that the tracker's attention can be properly restored when encountering tracking drift caused by occlusion or temporary scene distractions.

II. Related Work

A. Single Object Trackers

As a fundamental task in computer vision, SOT has attracted great research attention for its significance in various practical fields, such as video surveillance, automatic driving, robot navigation systems, and so on. Given the target's ground-truth in the first frame, which is usually presented as an object bounding box, SOT aims at predicting target states in the subsequent frames, including position and size.

To achieve SOT in generic videos, representative tracking algorithms can be categorized into the following three classes: CF-based methods, including non-CNN ones with hand-crafted features and fully CNN ones with deep learning features, Siamese network methods with fully CNN structures, Transformer-based methods involving CNN-Transformer ones and fully Transformer ones.

Starting with the seminal work of MOSSE [33], researchers designed a series of CF-based methods built on the discriminative solution that trains a specific classifier to distinguish the target from the background. They learn the optimal template filter online and apply a simple Hadamard product in the frequency domain through bi-directional Fourier transform to efficiently calculate feature correlation. Based on the mentioned model mechanism, these CF-based trackers mainly innovate from feature expression [14], [16], optimization approach [34], [35], overall learning objectives [15], [17], and target state estimate [16]. For instance, in order to enrich feature's information content, the exploited image features have evolved from pixel-level gray feature [33], [36], to artificially designed features like histogram of oriented gradient (HOG) [34] and color names (CN) [14], then to deep convolutional feature [14], [16], [37]. Besides, KCF [34] enlarges the training dataset with translated samples by carrying cyclic shifts and maps ridge regressions to more powerful ones in the nonlinear feature space with kernelized CF. ECO

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[14] takes principal components analysis (PCA) to decrease feature channels for lightweight filter learning and Gaussian mixture model (GMM) to generate representative components in training samples, achieving lower model complexity and enhanced real-time competence. ATOM [16] devises a target-oriented intersection over union (IoU) prediction module to assist the DCF-based classifier in selecting the final bounding box from rough estimates. Considering the expensive labeling cost of CNN's training samples, ALT [37] proposes a multi-frame cooperation-based active learning algorithm to select a diverse subset from massive candidates. To ensure the subsets' representativeness, it utilizes a nearest neighbor discrimination method to remove certain isolated or poor-quality sequences.

Regarding object tracking as a similarity learning problem, Siamese network methods pioneered by SiamFC [38] generate a target position response map by matching the similarity between SRs and the target template. These trackers [11], [12], [13], [38], [39], [40], [41] use Siamese CNNs with the same structure and shared parameters for two-branch feature transformation first, and then carry cross correlation operation for similarity computation. Concerning the inflexible multiscale test, SiamFC++ [12] adds a target-state regression branch and a quality assessment branch to refine estimation. In contrast with the anchor-free SiamFC [37], SiamRPN [39] brings the region proposal network (RPN) from object detection to embed multiaspect-ratio searching with anchors, achieving simultaneous classification and proposal regression for higher accuracy of bounding box prediction. To break through the restriction of translation invariance, SiamRPN++ [11] devises spatial-aware sampling for deeper CNNs like ResNet [42], channel-wise correlation for parameter balance, and multilevel aggregation for richer image perception. Since the optimization of CNN feature extractor requires effective constraints, the self-supervised method self-SDCT [43] constructs a multicycle consistency loss and a cycle trajectory consistency loss based on response-maps and pseudo-labels in forward–backward prediction, to align forward and backward tracking for improved accuracy and increase differences between adjacent frames for background suppression.

Recently, CNN-Transformer trackers [20], [21], [23], [25] and fully Transformer trackers [26], [27], [28], [29], [30], [32] are gradually becoming mainstream. Since these Transformer-based tracking methods have been discussed in detail in Section I, relevant contents are not repeated here.

B. Satellite Video Object Trackers

Consistent with the framework development in SOT, recent algorithms designed for SVOT can also be divided into groups of CF-based trackers [44], [45], [46], Siamese network trackers [6], [47], [48], CNN-Transformer trackers [4], [5] and fully Transformer trackers [2], [3].

Taking representative algorithms for example, CF-based CFME [44] advances the KCF [34] pipeline by proposing a Kalman filter-based motion estimation module with moving trajectory modeling, aiming to mitigate the object occlusion problem and KCF's boundary effect. However, the tracker is not flexible enough to tackle the distractions of similar surroundings. To build a robust model, Siamese-network HRSiam [6] adaptively fuses complementary advantages of tracking and detection by virtue of tracking deformation monitoring. Aiming at precise localization for typically small objects, Shao et al. [6] design a lightweight parallel network oriented to high spatial resolutions under the SiamRPN [39] framework. To cope with environmental interferences, the tracker employs interframe motion cues to refine primary tracking results by means of motion-based GMM [49] detection and Mean Shift [50]. However, its employed traditional detection algorithms may fail to handle large target variation. Faced with the object appearance deficiency challenge, CNN-Transformer TATrans [5] develops a bi-directional propagation and fusion strategy to attentively integrate multiscale features for small objects, and a target-aware enhancement module where target query is decoupled into content-aware and spatial-aware embeddings following conditional DETR [51], on the basis of the STARK [23] pipeline. However, the tracker neglects temporal information, thus causing incomplete target understanding. Fully Transformer TransMPP [3] adopts the OSTrack [26] pipeline to achieve the main one-stream tracking. To enable satellite-specific temporal modeling, Zhao et al. [3] utilize an attention mechanism to propagate short-term features between adjacent frames and update state features across long-term paths, producing more accurate classification and regression. To exploit global movement consistency, the tracker uses multivariate attention [52] for target state prediction based on the historical frame sequence. However, the whole model's computational complexity seems high with more learnable parameters and an overfitting risk.

In this article, we choose the fully Transformer framework to take advantage of the Transformer's long-term sequence modeling [19], and focus on the property of temporal sequence in satellite videos to fully exploit the underlying information of target changing tendency, robust target perception, and the target's continuous motion.

III. Proposed Approach

A. Overall Architecture

Fig. 1 illustrates the general procedure of our algorithm TSTrans, where three kinds of images, containing SR $i^{\mathrm{SR}} \in \mathbb{R}^{3\times256\times256}$, standard template $i^{\mathrm{ST}} \in \mathbb{R}^{3\times128\times128}$ and L temporal templates $i_k^{\mathrm{TT}} \in \mathbb{R}^{3\times128\times128}(k = 1,\ldots,L)$, are fed to experience feature engineering, long-term sequential target modeling, pruned tracking head, and motional correction in turn. It is worth mentioning that we expand the simple standard template i^{ST} to a long-term temporal template sequence $\{i^{\mathrm{ST}}, i_1^{\mathrm{TT}},\ldots,i_L^{\mathrm{TT}}\}$ in time order, different from the previous image-pair paradigm.

We follow the one-stream one-stage feature modeling based on the fully Transformer pipeline from OSTrack-256 [26], utilizing ViT-base [31] encoder as backbone for merged feature extraction and matching. Input images are cropped into 16×16 patches, containing $p^{\mathrm{ST}} \in \mathbb{R}^{n\times3\times16\times16}, p^{\mathrm{TT}} \in \mathbb{R}^{L\times n\times3\times16\times16}, p^{\mathrm{SR}} \in \mathbb{R}^{m\times3\times16\times16}$ at $n = (128/16)^2 = 64, m = (256/16)^2 = 256$, and linearly projected into patch embeddings, respectively. Then, in 12 stacked Transformer blocks, concatenated tokens are further distilled and reweighted with

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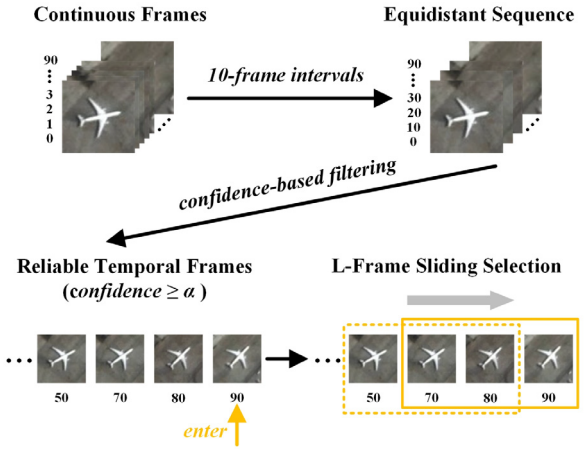


Fig. 2. Designed sliding selection for temporal templates.

intertoken attention, achieving feature enhancement and target-aware matching at the same time. More details about token design, temporally target-aware attention, and background pre-exclusion are discussed in Section III.

Given the target-aware SR feature map $f^{\mathrm{TA}} \in \mathbb{R}^{768\times16\times16}$, a pruned tracking head is applied to generate the predicted center (x^p, y^p) within SR. Subsequently, under motional supervision to relieve tracking drift, the center point coordinate is checked and conditionally corrected to the simulated motional result, computing the final tracked center (x^*, y^*). Owing to the material rigidity and moving simplicity of vehicles, airplanes, trains, or ships, objects of interest in satellite videos have almost invariant sizes and aspect ratios in the majority of cases. Thus, we directly displace the underperforming bounding box regression with a fixed size given in the initial frame.

B. Long-Term Sequential Target Modeling

Driven by temporal clues on the target's long-term variation pattern and multistage semantic features, our TSTrans improves the previous image-pair mode of $\{i^{\mathrm{ST}}, i^{\mathrm{SR}}\}$ to a chronologically arranged image sequence $\{i^{\mathrm{ST}}, i_1^{\mathrm{TT}}, \ldots, i_L^{\mathrm{TT}}, i^{\mathrm{SR}}\}$ covering standard template, L temporal templates and SR. Functioning as carriers of referenced target states, the selection of temporal templates is particularly important. According to two basic principles of reliability and temporally representativeness, we design an L-frame sliding selection mechanism. As shown in Fig. 2, we first determine a maximum selection frequency by setting checkpoints every ten frames for phased representativeness. Second, to guarantee template reliability, we control the accuracy of the cropped region by filtering out frames with low target-localizing confidence. Specifically, confidence is the maximum value in the center position map m^{CP} generated by tracking head with (8), and should be higher than the TT confidence threshold α. Third, we design a sliding window with a size of L and a stride of 1 to dynamically update temporal templates, once a new reliable frame passes a confidence-based checkpoint and enters the candidate sequence. Here, the sliding window is initialized with L copies of the standard template.

To demonstrate the advantages of our L-frame sliding selection mechanism, we have conducted a comparative experiment with a simpler selection strategy and even temporal spacing. Different from the unevenly sampled temporal templates brought by our confidence-based design, the comparative experiment first samples historical frames with a uniform interval of ten frames to form a candidate sequence. Whenever a new candidate enters the equidistant sequence (namely, every ten frames tracked), temporal templates are updated with the same sliding window of L-frame size and one-frame stride. Here, the setting of a uniform interval and a sliding window is consistent with our proposed mechanism for parallel comparison. As shown in Table I, the lower values of accuracy metrics in the last row (uniform TT) than our results in red font ($\alpha = 0.5$) indicate the benefits of our choice.

1) Token Design: Original patch tokens are composed of patch embeddings and position embeddings. Patch embeddings $e^{\mathrm{RGB}} \in \mathbb{R}^{p\times768}$ with hidden size of 1×768 are linearly projected and flattened from $3 \times 16 \times 16$ RGB-patches separately, where the number of patches p represents $m = 256$ for SR and $n = 64$ for templates. Specifically, the template or SR is first cropped and projected through a learnable convolutional layer with kernel size of 16×16, sliding stride of 16, input channel number of 3, and output channel number of 768. Then, 2-D feature maps are flattened to a set of 1-D embeddings. Position embeddings $e_T^{\mathrm{POS}} \in \mathbb{R}^{64\times768}$ or $e_{\mathrm{SR}}^{\mathrm{POS}} \in \mathbb{R}^{256\times768}$ are learnable parameters to indicate the absolute positions of different patches in the same image.

To identify tokens from standard template and temporal templates, we add two learnable ID embeddings $e_{\mathrm{ST}}^{\mathrm{ID}} \in \mathbb{R}^{64\times768}$ and $e_{\mathrm{TT}}^{\mathrm{ID}} \in \mathbb{R}^{64\times768}$ to the two kinds of template tokens $t^{\mathrm{ST}} \in \mathbb{R}^{64\times768}$ and $t^{\mathrm{TT}} \in \mathbb{R}^{L\times64\times768}$ respectively. Thus, our standard template patch tokens t^{ST} and temporal template patch tokens t^{TT} are computed as

$$t^{\mathrm{ST}} = e_{\mathrm{ST}}^{\mathrm{RGB}} + e_T^{\mathrm{POS}} + e_{\mathrm{ST}}^{\mathrm{ID}} \tag{1}$$

$$t^{\mathrm{TT}} = e_{\mathrm{TT}}^{\mathrm{RGB}} + e_T^{\mathrm{POS}} + e_{\mathrm{TT}}^{\mathrm{ID}}. \tag{2}$$

This discrimination of ID embeddings encourages the model to treat each kind of template tokens on its merits, in other words, to learn basic target appearance from standard templates and enrich its target-related knowledge with dynamic changes from sequential temporal templates. In order to demonstrate the necessity of ID embeddings $e_{\mathrm{ST}}^{\mathrm{ID}}$ and $e_{\mathrm{TT}}^{\mathrm{ID}}$, we have carried out an ablation experiment where ID embeddings are removed. As shown in Table II, compared with our results in the second last line, the reduction of three accuracy metrics in the last row (w/o ID) shows the positive effects of template ID embeddings. While two kinds of template tokens are identified in this way, SR tokens $t^{\mathrm{SR}} \in \mathbb{R}^{256\times768}$ involve no ID embedding. The reason is that position embeddings e_T^{POS} and $e_{\mathrm{SR}}^{\mathrm{POS}}$ already carry learned information for identifying template tokens and SR tokens. Thus, SR patch tokens are calculated as

$$t^{\mathrm{SR}} = e_{\mathrm{SR}}^{\mathrm{RGB}} + e_{\mathrm{SR}}^{\mathrm{POS}}. \tag{3}$$

2) Temporally Target-Aware Attention: Fig. 3 demonstrates the structure of key blocks in our modified ViT-base [31] backbone. Such blocks are stacked 12 times, and only 3 of them are equipped with "background pre-exclusion." In

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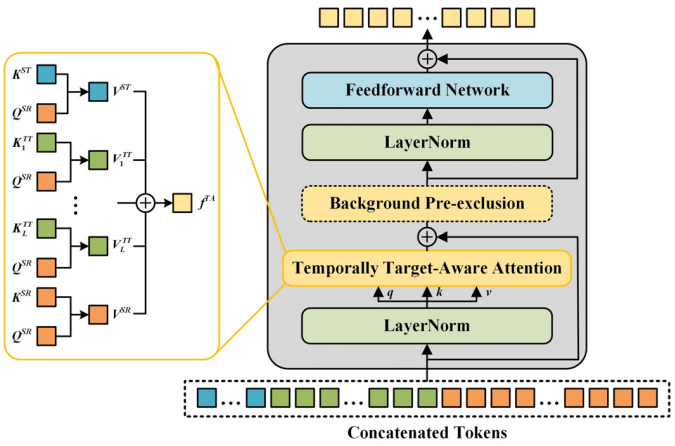


Fig. 3. Detailed inner structure of the basic block in the backbone.

our temporally target-aware attention, three kinds of patch tokens $t^{\mathrm{ST}}, t^{\mathrm{TT}}, t^{\mathrm{SR}}$ are first concatenated in p dimension and transformed to corresponding vectors of query, key, and value by learnable mapping matrices. Then, based on the mechanism of the multihead self-attention, dense intertoken relationships are modeled as (4), shown at the bottom of the page, where $[\cdots;\cdots]$ infers token concatenation operator, A_q^k denotes attention coefficient matrix calculated on query vector from q and key vector from k. For template tokens, $A_{iT}^{iT}V^{iT}(i \in \{S, T(1), \ldots, T(L)\})$ extracts intratemplate dependency to enhance feature representativeness, $A_{iT}^{jT}V^{jT}(i, j \in \{S, T(1), \ldots, T(L)\}; i \neq j)$ extracts intertemplate interactions to learn target changing tendency from a long-term temporal variation perspective, $A_{iT}^{\mathrm{SR}}V^{\mathrm{SR}}(i \in \{S, T(1), \ldots, T(L)\})$ adds contextual information from the current frame to emphasize target orientation. For SR tokens, $A_{\mathrm{SR}}^{\mathrm{SR}}V^{\mathrm{SR}}$ makes further feature extraction, $A_{\mathrm{SR}}^{iT}V^{iT}(i \in \{S, T(1), \ldots, T(L)\})$ aggregates target-aware similarities between SR patch and multiple sequential target states to develop complete and robust object understanding. Therefore, feature enhancement and target-aware matching are progressively realized block by block. Finally, the target-aware SR feature is generated by maintaining the output SR tokens and reshaping them to a 2-D form as

$$\begin{aligned} f^{\mathrm{TA}} &= 2\mathrm{D}\left(A_{\mathrm{SR}}^{\mathrm{ST}}V^{\mathrm{ST}} + A_{\mathrm{SR}}^{\mathrm{TT}(1)}V_1^{\mathrm{TT}} + \cdots + A_{\mathrm{SR}}^{\mathrm{TT}(L)}V_L^{\mathrm{TT}} + A_{\mathrm{SR}}^{\mathrm{SR}}V^{\mathrm{SR}}\right) \\ &= 2\mathrm{D}\left(f_{\mathrm{SR}}^{\mathrm{ST}} + f_{\mathrm{SR}}^{\mathrm{TT}(1)} + \cdots + f_{\mathrm{SR}}^{\mathrm{TT}(L)} + f_{\mathrm{SR}}^{\mathrm{SR}}\right) \end{aligned} \tag{5}$$

where $2\mathrm{D}(\cdot)$ infers 1-D–2-D reshaping operator, and f_q^{kv} denotes refined feature from q with correlation between features from kv and q. f^{TA} integrates not only spatial information from intraframe dependencies and template-frame similarities, but also temporal cues from dense relationship modeling between SR and long-term template sequence.

3) Background Pre-Exclusion: We keep the "early candidate elimination" module of OSTrack [26], where in the fourth, seventh, and tenth blocks, low-scoring (bottom 0.3 of the score ranking) patch tokens from SR are assigned to zero vectors for progressive exclusion of suspected background patches, contributing to lower computational cost and less distraction. To adapt to more references from temporal templates, we expand the original score to total intertoken similarities between a certain SR token and all template center tokens

$$\begin{aligned} \mathrm{score} &= \mathrm{Softmax}\left(\frac{Q_c^{\mathrm{ST}}\left[K^{\mathrm{SR}}\right]^{\mathrm{T}}}{\sqrt{d}}\right) \\ &\quad + \mathrm{Softmax}\left(\frac{Q_c^{\mathrm{TT}(1)}\left[K^{\mathrm{SR}}\right]^{\mathrm{T}}}{\sqrt{d}}\right) \\ &\quad + \cdots + \mathrm{Softmax}\left(\frac{Q_c^{\mathrm{TT}(L)}\left[K^{\mathrm{SR}}\right]^{\mathrm{T}}}{\sqrt{d}}\right) \end{aligned} \tag{6}$$

where $Q_c^{iT}(i \in \{S, T\})$ represents the query vector of the center token in the template labeled as iT. In this way, the similarity score is more reliable for discriminating between target candidates and background tokens, with a more comprehensive reference from the whole temporal template sequence.

4) Module Advantages: With regard to attention modeling, TSTrans's temporally target-aware attention differs from the attention mechanisms in existing Transformer-based trackers that utilize temporal information: Most of them take OSTrack [26] as a baseline, where only the fixed first frame is selected as the target-aware reference. The previous simplistic temporal approach adds a single DT, capturing only the latest target change without forming a sequential temporal perspective. SeqTrack [27] converts spatial heatmap prediction into sequential bounding box values, which does not incorporate concepts of sequence into target-aware references or visual encoder features; instead still uses a static template (ST) like OSTrack [26] or combines ST with the single DT. Although ProContEXT [29] employs multiple templates, it supplements target-aware references from the spatial dimension. For the ST or DT, it observes the target using two distinct zoom ratios

$$\begin{aligned} \mathrm{TTA-Attention} &= \mathrm{Softmax}\left(\frac{\left[Q^{\mathrm{ST}}; Q_1^{\mathrm{TT}}, \ldots, Q_L^{\mathrm{TT}}; Q^{\mathrm{SR}}\right]\left[K^{\mathrm{ST}}; K_1^{\mathrm{TT}}, \ldots, K_L^{\mathrm{TT}}; K^{\mathrm{SR}}\right]^{\mathrm{T}}}{\sqrt{d}}\right) \\ &\quad \cdot \left[V^{\mathrm{ST}}; V_1^{\mathrm{TT}}, \ldots, V_L^{\mathrm{TT}}; V^{\mathrm{SR}}\right] \\ &= \begin{pmatrix} A_{\mathrm{ST}}^{\mathrm{ST}} & A_{\mathrm{ST}}^{\mathrm{TT}(1)} & \cdots & A_{\mathrm{ST}}^{\mathrm{TT}(L)} & A_{\mathrm{ST}}^{\mathrm{SR}} \\ A_{\mathrm{TT}(1)}^{\mathrm{ST}} & A_{\mathrm{TT}(1)}^{\mathrm{TT}(1)} & \cdots & A_{\mathrm{TT}(1)}^{\mathrm{TT}(L)} & A_{\mathrm{TT}(1)}^{\mathrm{SR}} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ A_{\mathrm{TT}(L)}^{\mathrm{ST}} & A_{\mathrm{TT}(L)}^{\mathrm{TT}(1)} & \cdots & A_{\mathrm{TT}(L)}^{\mathrm{TT}(L)} & A_{\mathrm{TT}(L)}^{\mathrm{SR}} \\ A_{\mathrm{SR}}^{\mathrm{ST}} & A_{\mathrm{SR}}^{\mathrm{TT}(1)} & \cdots & A_{\mathrm{SR}}^{\mathrm{TT}(L)} & A_{\mathrm{SR}}^{\mathrm{SR}} \end{pmatrix} \begin{pmatrix} V^{\mathrm{ST}} \\ V_1^{\mathrm{TT}} \\ \vdots \\ V_L^{\mathrm{TT}} \\ V^{\mathrm{SR}} \end{pmatrix} \end{aligned} \tag{4}$$

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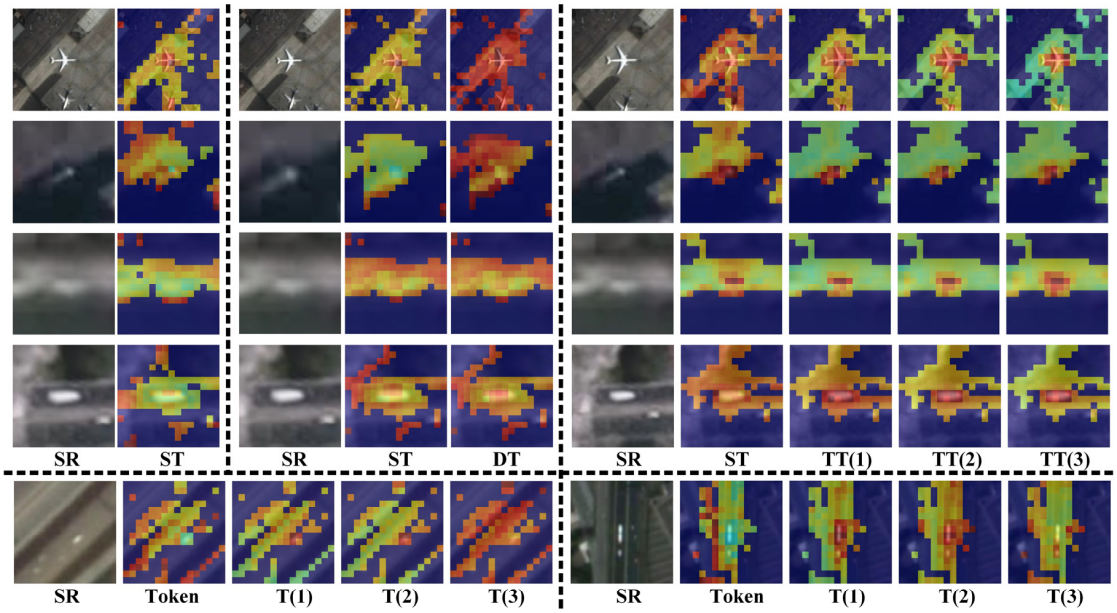


Fig. 4. Visualization of attention maps. (Row 1–4: OSTrack [26], OSTrack [26]-ST&DT, and TSTrans|(from left to right) Row 5: ODTrack [30].)

while maintaining the same image resolution, resulting in two contexts of differing sizes. In contrast, TSTrans expands target-aware references from the temporal dimension by selecting multiple key frames to represent historical frame sequence, thereby extracting long-term sequential information such as target changing trendency and enhanced multitemporal perception. ODTrack [30] primarily proposes a single temporal token to accumulate motion or global target-related information from historical frames, and simply takes equidistant frames as reference, lacking sequential representativeness. LMTrack [32] converts the form of a single target-aware reference from a full template region to an individual template patch, utilizing a cache to store the top important template tokens based on attention scores and classification contribution.

To quantitatively measure the benefits brought by our proposed module, we design a metric to indicate the target-aware information capacity within an attention mechanism. In a high-capacity mechanism, the attention map between reference and SR should focus on the target's location and shape, approaching the ideal distribution of target-background probability. Therefore, the calculation method for this metric is as follows. For the attention scores generated in the final stage of feature extraction backbone, we first compute the average attention map between all patch tokens within each reference and the overall SR patch tokens (Q from SR, K from the reference), then aggregate all references. Next, a Gaussian map of the target in SR $M^{\mathrm{gt}} \in \mathbb{R}^{16\times16}$ is generated based on the ground-truth bounding box. Finally, the metric is calculated as focal loss [53] between these two maps. A smaller loss value denotes higher target-related accuracy in the attention map, signifying better module performance. Taking TSTrans as an example, the calculation for this metric can be formed as

$$\text{Capacity} = L_{\text{focal}}\left(\sum_i A_{\text{SR}}^{iT}, M^{\text{gt}}\right), \quad i \in \{S, T(1), \ldots, T(L)\}. \tag{7}$$

TABLE I
TT CONFIDENCE THRESHOLD α ADJUSTMENT

Dataset	Methods	AUC-P (%)	AUC-S (%)	Prec-5 (%)
SV248S	$\alpha = 0.1$	81.33	38.61	76.98
	$\alpha = 0.2$	81.81	38.80	77.20
	$\alpha = 0.3$	81.09	38.21	75.21
	$\alpha = 0.4$	81.28	38.81	75.83
	$\boldsymbol{\alpha = 0.5}$	**86.36**	**40.84**	**80.62**
	$\alpha = 0.6$	82.06	37.41	73.58
	$\alpha = 0.7$	82.51	37.79	73.58
	$\alpha = 0.8$	82.96	38.16	73.73
	$\alpha = 0.9$	82.13	38.78	75.39
	uniform TT	81.23	38.88	77.47

On the SatSOT [8] test set, we compute the mean value of this metric across all tested frames. For cases where M^{gt} is all-zero (improper SR caused by tracking failure), we use zero-filled attention maps to calculate meaningful loss values. Table III shows the comparison among OSTrack [26], OSTrack-ST&DT [26], ProContEXT [29], ODTrack [30], LMTrack [32], and TSTrans, where our TSTrans achieves the best performance. ProContEXT [29] introduces certain background noise when expanding the template's context. Some learned temporal tokens of ODTrack [30] in the very early stage of videos have low relevance to targets, even disturbing target matching in other templates, as shown in the last row in Fig. 4. LMTrack [32], when adopting independent template tokens, disrupts templates' inner spatial arrangement, losing certain useful spatial information.

As shown in Fig. 4, we visualize the attention maps of each target-aware reference during tracking. Compared to our baseline OSTrack [26] and its version incorporating single DT (OSTrack-ST&DT), the high-score regions in attention maps of our TSTrans are more concentrated on the target, with more appropriate probability distributions. In the second row, when

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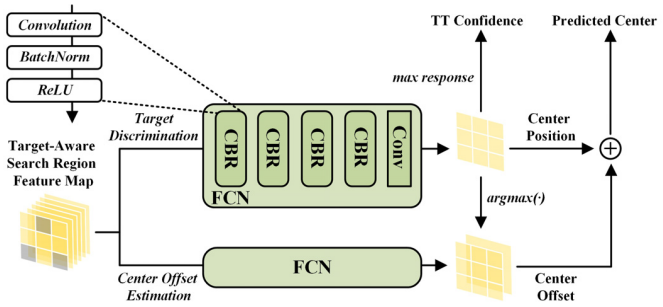


Fig. 5. Framework of pruned tracking head.

the target is occluded by shadows, TSTrans remains focused on the target location while other trackers' attentions are distracted. In the third row, when a similar object appears around the target and the target's appearance is blurred, TSTrans still assigns higher attention scores and accurate coverage to the tracked vehicle, while other trackers fail to distinguish the target from the background. In the fourth row, when background elements have a similar color or texture to the target, TSTrans effectively suppresses confusing background parts, whereas other trackers are disrupted.

Furthermore, the visualization in Fig. 4 reveals the distinct contributions of different temporal templates in TSTrans. Taking the first row as an example, as the time point approaches the present, patches with high attention scores progressively reduce to the target region. Besides, the standard template typically highlights more areas where the target might exist, serving as a supplementary role.

C. Tracking Head With Global Refinement

In this section, we introduce the latter part of TSTrans, a center point predictor as our tracking head, and a motional correction module as a global refinement of predicted results.

1) Center Point Predictor: In the testing phase of one-pass evaluation (OPE), the current frame's searching view is usually determined based on the last frame's target size prediction. Therefore, inaccurate size regression and its associated error accumulation can have negative impacts on tracking performance. Considering the targets' shape steadiness and trackers' poor regression ability, we prune the common three-branch tracking head by cutting the bounding box regression branch to more concentratedly localize the target center.

As shown in Fig. 5, we employ the remaining target discrimination branch $F^{\mathrm{cp}}(\cdot)$ to generate the center position's localizing response map $m^{\mathrm{CP}} \in \mathbb{R}^{1\times16\times16}$ and the center offset estimation branch $F^{\mathrm{co}}(\cdot)$ to produce the center coordinate's offset response maps $m^{\mathrm{CO}} \in \mathbb{R}^{2\times16\times16}$. The two branches have independent network parameters and the same structure, which consists of four 3×3 convolutional blocks and one 1×1 convolutional layer. To decode the coordinate of the predicted center, we first find the maximum response in m^{CP} as current frame's TT confidence for updating temporal templates, and its position (X^p, Y^p) as the basic center coordinate. Then, the corresponding responses at position (X^p, Y^p) in m^{CO} are added to make up local offsets

$$m^{\mathrm{CP}} = F^{\mathrm{cp}}\left(f^{\mathrm{TA}}\right) \tag{8}$$

TABLE II
ABLATION EXPERIMENTS RESULTS

Dataset	Methods			AUC-P (%)	AUC-S (%)	Prec-5 (%)
	a	*b*	*c*			
SatSOT				65.87	46.59	**57.91**
	✓			**69.66**	**50.22**	**57.66**
		✓		64.84	42.87	55.61
			✓	63.81	46.48	55.30
	✓	✓		64.46	43.91	54.98
	✓		✓	**67.78**	**52.32**	56.51
		✓	✓	66.42	47.27	56.67
	✓	✓	✓	**71.00**	**53.20**	**61.43**
	w/o ID			69.24	52.41	60.13

$$m^{\mathrm{CO}} = F^{\mathrm{co}}\left(f^{\mathrm{TA}}\right) \tag{9}$$

$$(X^p, Y^p) = \underset{X,Y\in\{1,2,\ldots,16\}}{\arg\max}\left(m^{\mathrm{CP}}[X][Y]\right) \tag{10}$$

$$(x^p, y^p) = (X^p, Y^p) + m^{\mathrm{CO}}\left[X^p\right]\left[Y^p\right] \tag{11}$$

where $m[i][j]$ infers taking the value at (i, j) in map m.

For training losses, we adopt the L1 difference between predicted center and ground truth center as L_{l1} to supervise the whole network involving the backbone and two-branch head. Meanwhile, the focal loss [53] L_{focal} about the center position map and ground truth center heatmap is utilized to further constrain the backbone and target discrimination branch. The total loss L_{total} is a weighted sum of the two losses

$$L_{\mathrm{total}} = \mu_1 L_{l1} + \mu_2 L_{\mathrm{focal}} \tag{12}$$

where we set $\mu_1 = 5$ and $\mu_2 = 1$.

To validate the effectiveness of fixing the bounding box size and eliminating its regression branch, we have conducted three comparative experiments. These experiments, respectively, predict the bounding box's width and height, aspect ratio multiplier (ratio of the first frame's bounding box aspect ratio to the current target's aspect ratio), and scale multiplier (ratio of the first frame's bounding box area to the current target's bounding box area), with an extra regression branch. For [Ours + (w/h)] and [Ours + (w^*h)], we adjust the bounding box size with the predicted multiplier, with respective assumptions of unchanged area and aspect ratio. As shown in Table IV, although our approach reduces the flexibility of predicted bounding box shape, it enhances the model's accuracy in locating targets with more fit model parameters, and prevents the error accumulation caused by incorrect size regressions.

2) Motional Correction Module: Driven by the target's moving continuity from historical frame sequence, we build a motional correction module to cope with tracking drift caused by occlusion and distraction. For example, a targeted black vehicle will be confused with other vehicles or a dark road patch, if the target gets occluded, or a similar object appears in tracking view. Thus, this module detects trajectory departure and timely recovers tracker attention to the proper visual field, enlightened by SiamMDM [48]. To simulate targets' movements, we utilize the conventional assumption of short-term uniformly linear motion [44], on account of short

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Algorithm 1 Motional Correction Module

Input:

(1) t, id of the current search frame which starts from 1, set $N = t$ at $t < \beta$ and $N = \beta$ at $t \geq \beta$;
(2) $c^*_{t-1} = (x^*_{t-1}, y^*_{t-1})$, the last tracked center;
(3) $c^p_t = \{(x^p_{t-N+1}, y^p_{t-N+1}), \ldots, (x^p_t, y^p_t)\}$, dynamic cache of historical predicted centers;
(4) $\Delta^G = \{(\Delta x^p_1, \Delta y^p_1) \ldots (\Delta x^p_{t-N+2}, \Delta y^p_{t-N+2}) \ldots (\Delta x^p_{t-1}, \Delta y^p_{t-1})\}$, global cache of historical average displacements in x and y directions.

Output:

$c^*_t = (x^*_t, y^*_t)$, the current tracked center.

length of existing reference: $N \leftarrow \text{len}(c^p_t)$;
if $N = 1$ **then**
  $\Delta x^p_t \leftarrow 0, \Delta y^p_t \leftarrow 0$:
else
  current average displacements: $(\Delta x^p_t, \Delta y^p_t) \leftarrow$ Eq. (13, 14);
end if
$\Delta^G \leftarrow \Delta^G \cup \{(\Delta x^p_t, \Delta y^p_t)\}$;
if $t = 1$ or $t = 2$ **then**
  $(x^m_t, y^m_t) \leftarrow (x^p_t, y^p_t)$;
else
  ideal movements: $(x^\Delta_t, y^\Delta_t) \leftarrow$ Eq. (15, 16);
  motional center: $(x^m_t, y^m_t) \leftarrow (x^p_t, y^p_t) + (x^\Delta_t, y^\Delta_t)$;
end if
if Distance $[(x^p_t, y^p_t), (x^m_t, y^n_t)] < \gamma$ **then**
  keep predicted center as tracked: $c^*_t \leftarrow (x^p_t, y^p_t)$;
else
  take motional center as tracked: $c^*_t \leftarrow (x^m_t, y^m_t)$.
end if

interframe time interval, small displacement, and unvarying moving direction in a very short-range period.

Referring to Algorithm 1, the whole procedure with details is presented in pseudo-code. First, we record the predicted center ranging from the past $(\beta - 1)$th frame to the current frame, where β denotes the historical reference's length. Here, we choose the predicted center as reference rather than the motional center or final result, in order to avoid a pure linear trajectory and reduce error accumulation by introducing more visual adaptiveness. Second, long-term displacement instead of adjacent distance is averaged for every historical time node. So we initialize the first item as $(0, 0)$, and regularly update the global cache Δ^G with the newest item calculated as follows:

$$\Delta x^p_t = \left(x^p_t - x^p_{t-N+1}\right) / (N-1) \tag{13}$$

$$\Delta y^p_t = \left(y^p_t - y^p_{t-N+1}\right) / (N-1). \tag{14}$$

Third, we evenly combine all average displacements corresponding to the past $(N-1)$ historical frames to generate the ideal movement for the current frame

$$x^\Delta_t = \sum_{i=t-N+2}^{t} \Delta x^p_i / (N-1) \tag{15}$$

$$y^\Delta_t = \sum_{i=t-N+2}^{t} \Delta y^p_i / (N-1). \tag{16}$$

Finally, we obtain the motional center $c^M_t = (x^m_t, y^m_t)$ with the ideal distance from its last tracked center c^*_{t-1}.

To evaluate the degree of trajectory departure, in other words, the extent of attention deviation, we apply a checking mechanism based on the difference between the predicted center and motional center. Once the Euclidean distance between the two points exceeds the attention distraction threshold γ (pixels), we correct the predicted center to the motional one as the final result.

IV. Evaluation

A. Experimental Setup

We conduct model training and testing on the following three satellite video datasets labeled with SOT. To test different trackers on all three datasets, respectively, we train three versions of trackers on the SatSOT [8] training set, SV248S [9] training set, and VISO [10] training set, separately.

1) *SatSOT:* This dataset contains 105 video sequences collected by Jilin-1, Skybox, and Carbonite-2. The video lengths range from 120 frames to 750 frames, with an average of 263 frames per video. The targets include 65 cars, 26 trains, 9 airplanes, and 5 ships, within images of 12 000 × 5000 pixels. It features large spatial scales, complex traffic scenarios, and diverse target sizes varying from 21 to 780 605 pixels. We divide SatSOT [8] into a 79-sequence training set and a 26-sequence testing set in a ratio of 3:1. It is worth noting that the "train-test" data split ensures basically consistent proportions of four object categories in the two sets, and no overlapping scenes between the two sets, which calls more for model generalization ability.
2) *SV248S:* This dataset includes 248 video sequences created from six videos in Chang Guang's open-source data. The numbers of frames in these videos range from 500 to 753. The targets contain 202 vehicles, 37 large vehicles, 6 airplanes, and 3 ships, within images of 3840 × 2160 to 4096 × 2160 pixels and a maximum spatial resolution of 0.92 m. It features various environments covering five cities from four countries, and tiny targets with vehicle bounding boxes' diagonal lengths averaging at 8.1 pixels. To guarantee no overlaps between training scenes and testing scenes, we select all 47 sequences from the one video captured at San Francisco, California, USA, to constitute the testing set. In this way, we divide SV248S [9] into a 201-sequence training set and a 47-sequence testing set in a ratio of 4:1.
3) *VISO:* This dataset contains 47 videos captured by Jilin-1. 3630 effective video sequences with 1000 × 1000 images in the dataset are utilized, involving 3618 cars, 8 airplanes, and 4 ships. It features abundant target instances, of which 90% reach less than 50 pixels, bringing a greater challenge of tracking environment diversity for model evaluation. Keeping principles of no overlapping scenes and basically consistent proportions of three object categories for the "train-test" data split, we divide VISO [10] into 2692-sequence training set

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from videos 1–20, 39, 42, 44, 45, and 938-sequence testing set from videos 21–27, 40, 43, 47, still in accordance with the ratio 3:1.

1) Implementation Details: We train TSTrans on four NVIDIA GeForce RTX 3090 GPUs with the PyTorch framework. Our network uses a ViT-base [31] backbone pretrained on ImageNet [54] with MAE [55]. TSTrans is trained for 600 epochs with eight samples per batch and AdamW [56] optimizer, where the weight decay is set to 1×10^{-4}, the learning rate is initialized as 4×10^{-5} for backbone and 4×10^{-4} for other parts. During training, learning rates are decreased by a factor of 10 and 100 after 240 and 480 epochs, respectively. We adopt brightness jittering, horizontal flip, grayscale transformation, and SR cropping jittering for data augmentation. On the basis of enough controlled experiments, we set the hyperparameters as follows: number of temporal templates $L = 3$, TT confidence threshold $\alpha = 0.5$, historical reference's length $\beta = 64$, attention distraction threshold $\gamma = 5$.

For training, the template is cropped in a square with an area 4 times that of the annotated bounding box and then resized to 128×128; SR is cropped in a square with an area 16 times that of the annotated bounding box and then resized to 256×256. For inference, the operating way and size remain the same, while the cropping references of temporal templates and SR change to historical tracking results. For each training epoch, we randomly sample a training video to acquire one image group, for 2400 (SatSOT, SV248S) or 2692 (VISO) times. According to the randomly chosen SR id id^{SR} and standard template id id^{ST} in each video, we sample one temporal template id from $[id^{\mathrm{SR}} - 10, id^{\mathrm{SR}})$ if $id^{\mathrm{ST}} < id^{\mathrm{SR}}$ and from $[id^{\mathrm{SR}}, id^{\mathrm{SR}} + 10)$ otherwise.

2) Evaluation Metrics: During inference, we choose the OPE way where every video sequence is initialized by the ground truth bounding box in the first frame and runs on the tracker for one time without interruption. The adopted three evaluation metrics are as below.

1) *Success Plot:* The ordinates of points on the success plot indicate the proportion of frames where IoU between the tracked bounding box and ground truth is higher than the corresponding abscissa. The success plot's abscissa axis presents different overlap thresholds from a range of 0–1. To assess the tracker's overall performance on the success plot with a numerical metric, we set AUC-S by calculating the area under the success curve.
2) *Precision Plot:* The ordinates of points on the precision plot indicate the proportion of frames where the Euclidean distance between the tracked center and ground truth is lower than the corresponding abscissa. Precision plot's abscissa axis presents different location error thresholds from a range of 0–50 pixels. To fully describe the tracker's performance on the precision plot, we set both AUC-P by computing the area under the precision curve, and Prec-5 by picking the point value when the location error threshold is 5 pixels, on account of small object sizes in satellite videos.
3) *OPE Speed:* In order to depict the tracker's inference speed in real-time OPE, we employ the FPS metric that calculates the average number of tracked frames during one second.

TABLE III
ATTENTION'S TARGET-AWARE INFORMATION CAPACITY

Trackers	Capacity	Trackers	Capacity
OSTrack	10.95	ODTrack	9.16
OSTrack-ST&DT	10.78	LMTrack	10.91
ProContEXT	9.57	**TSTrans**	**8.36**

B. Experiment 1 (Comparison With 12 SOTA Trackers)

1) Quantitative Results: In order to evaluate the performance of the proposed TSTrans, we conduct parallel experiments on three datasets involving SatSOT [8], SV248S [9] and VISO [10], and comprehensively contrast our method with 12 SOTA tracking algorithms, comprising CF-based methods (KCF-2015-TPAMI [34], ECO-2017-CVPR [14], and STRCF-2018-CVPR [15]), satellite video oriented motion-assisted method CFME-2020-TGRS [44], Siamese network methods (SiamRPN++-2019-CVPR [11] and SiamFC++-2020-AAAI [12]), CNN-Transformer methods (STARK-2021-ICCV [23]) and fully Transformer methods (OSTrack-2022-ECCV [26], ProContEXT-2023-ICASSP [29], SeqTrack-2023-CVPR [27], ODTrack-2024-AAAI [30], and LMTrack-2025-AAAI [32]). All trackers are trained and tested on the same training and testing sets discussed in "data description" of Section IV. For parallel comparison, we retrain all trackers on the satellite video datasets under identical conditions instead of using public checkpoints. Regarding the five fully Transformer trackers, we choose their versions with 256×256 SR, for parallel comparison with TSTrans. Quantitative comparisons with results of AUC-P, AUC-S, Prec-5, and FPS are illustrated in Table V, according to which we make the analysis below.

Regarding SatSOT [8], our TSTrans ranks first on the three accuracy metrics, achieving improvements of 3.68% AUC-P, 4.76% AUC-S, and 1.59% Prec-5 over the second-highest metric values from LMTrack [32] and CFME [44]. With the baseline reaching AUC-P of 65.87% and AUC-S of 46.59%, our method achieves a much higher AUC-P of 71.00% and AUC-S of 53.2%. In terms of SV248S [9], our TSTrans realizes the top performance with AUC-P of 86.36% and AUC-S of 40.84%, representing accuracy gains of 6.86%/12.11% in AUC-P and 3.9%/8.01% in AUC-S, contrasted with the second ranking metrics from STRCF [15] or SiamFC++ [12], and the baseline, respectively. As for VISO [10], our TSTrans surpasses all the listed advanced trackers with scores of 65.38% in AUC-P, 28.05% in AUC-S, and 51.80% in Prec-5, outperforming the second ranking STRCF [15] and the baseline by 1.96%/16.34%, 0.73%/7.93%, and 0.89%/12.43% in AUC-P, AUC-S, and Prec-5, separately. It can be observed that the CF-based tracker STRCF [15] performs well and is close to the best on VISO [10]. This phenomenon may be attributed to the harder challenge of model generalization brought by more diverse tracking scenes and plentiful testing sequences in VISO [10], to which the target-specific template learning in CF-based approaches naturally leads to better coping capacity. Despite

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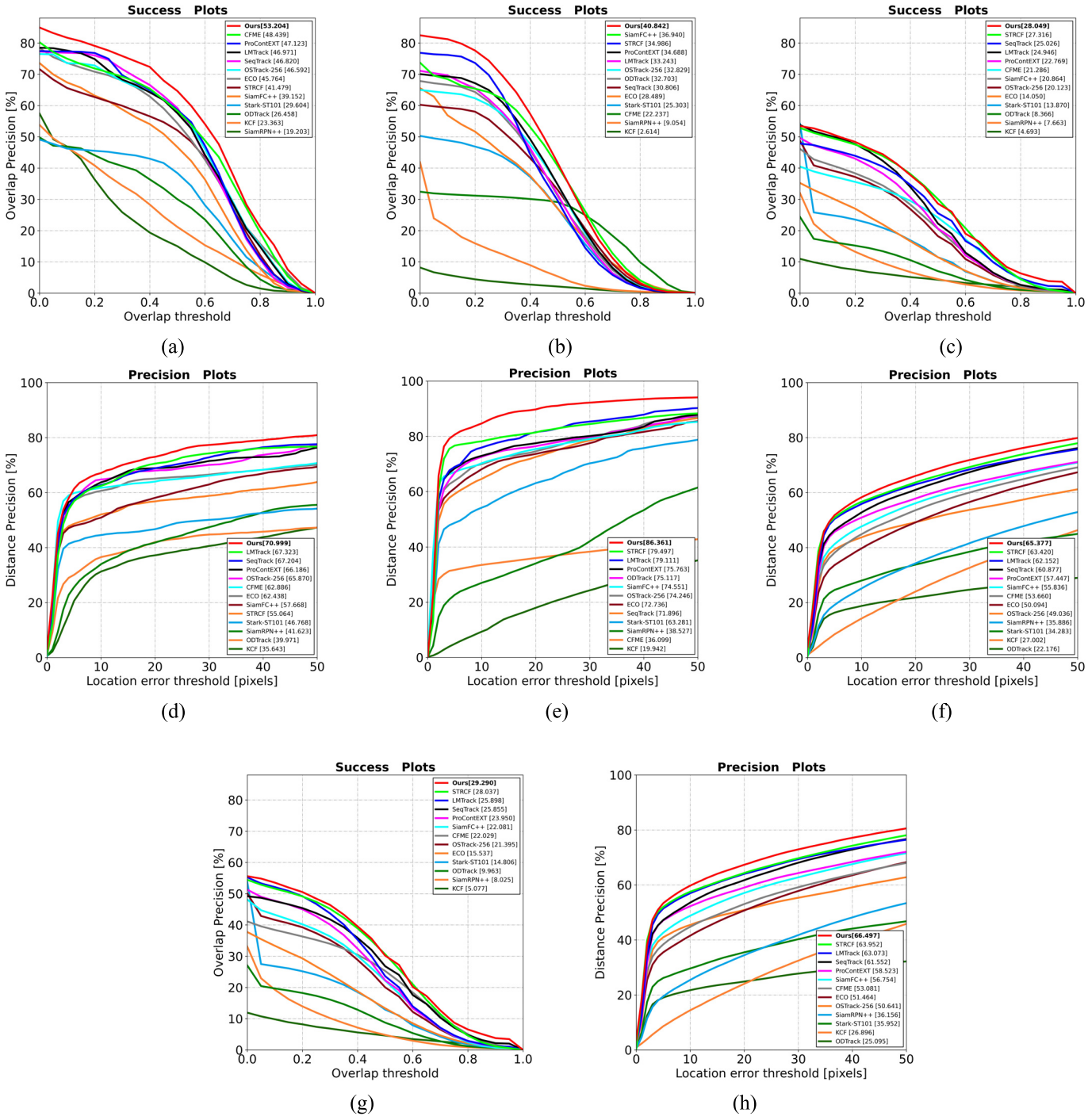


Fig. 6. Success plots and precision plots of all trackers on SatSOT [8], SV248S [9], VISO [10], and the overall testing set. The values in legends are AUC-S and AUC-P for success plots and precision plots separately. Success plots on (a) SatSOT, (b) SV248S, and (c) VISO. Precision plots on (d) SatSOT, (e) SV248S, and (f) VISO. (g) Success plots on three testing sets. (h) Precision plots on three testing sets.

this, the proposed TSTrans is more robust than other trackers, for it reaches higher overlap precision when the threshold becomes stricter, as proven by Fig. 6(c). In addition, for overall performance on the three datasets, we put together the inference results on three testing sets, and compute the three accuracy metrics as well as average FPS on them. From this perspective, the proposed method still performs superiorly with a top AUC-P of 66.50% and the best AUC-S of 29.29%. Here, the overall results are greatly influenced by tracker performances on VISO [10], because of a large amount of testing data in VISO [10].

In Table V, we can observe that ODTrack [30] performs at upper middle level in SV248S [9] but worse than other Fully Transformer algorithms in SatSOT [8] and VISO [10]. To analyze the reason, we find that the fit ODTrack models for the two datasets are inclined to generate false target locations and, more importantly, oversized bounding boxes, at the very early stage of some video sequences, bringing error accumulation

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TABLE IV
DIFFERENT COMPUTATIONS OF BOUNDING BOX SIZE

Dataset	Methods	AUC-P	AUC-S	Prec-5
SatSOT	Ours + (w, h)	64.46	43.91	54.98
	Ours + (w/h)	69.59	50.56	63.73
	Ours + (w*h)	70.45	51.08	58.11
	Ours	**71.00**	**53.20**	**61.43**

and improperly enlarged visual range, which directly cause the poor performance. Although the temporal token proposed by ODTrack [30] captures target-related information, it is initialized with a fixed token in each video sequence and propagated by adding the current temporal token to the fixed token for the use of the next frame's attention modeling. As a learnable parameter, this fixed token may learn some distributions of target traits in the dataset during training, like the average target size. But object sizes in SatSOT [8] or VISO [10] vary greatly with a high dispersion of distribution, which leads to low representativeness of the learned fixed token and its failure in adapting to targets with different sizes.

For speed assessment, our TSTrans belongs to the upper middle level, running at around 50 FPS, which achieves real-time inference. The faster trackers SiamFC++ [12] and KCF [34] can run at over 100 FPS, which is because of SiamFC++'s simpler model architecture, and KCF's accelerated designs of rapid correlation calculation and cyclic shift. It is noted that CF-based trackers are tested on cpu.

As shown in Table VI, regarding complexity comparison, we calculate multiply accumulate operations (MACs) of the main models in different algorithms as computational complexity, and count the sizes of their main trainable parameters (Params) as space complexity. Compared to the baseline [26], our template sequence mechanism does not significantly increase computational complexity. And our branch pruning leads to decreased space complexity. Our overall complexity comes at a moderate level in Transformer-based models.

Fig. 6 displays success plots and precision plots of all algorithms on the separate three testing sets and their collection, where our TSTrans's curves are always distributed at the outermost, presenting SOTA accuracy. In a word, the proposed fully Transformer model TSTrans realizes a competitive performance and a satisfactory balance between effectiveness and efficiency for SVOT.

2) Qualitative Results: For qualitative comparisons, we visualize the tracked bounding boxes of our proposed TSTrans along with six tracking algorithms performing relatively well: CF-based trackers STRCF [15] and CFME [44], Siamese network tracker SiamFC++ [12], fully Transformer trackers OSTrack [26], SeqTrack [27], and ProContEXT [29]. Seven challenging video sequences, including target categories of vehicle, train, plane, and ship, are selected for visual presentation in Fig. 7, where results from various trackers are distinguished with different colors.

In the second and seventh rows, typical situations of full occlusion impede other trackers from continuous localization, while TSTrans maintains accurate tracking in a long range, even at the 375th or 322nd frame. In addition, obvious illumination variation can be observed in the second row, where TSTrans successfully resists this kind of noise. In the first, third, and fourth rows, other trackers are easily confused by similar vehicle neighbors, while TSTrans greatly alleviates this problem. In the last three rows, mistaken background components and improperly oversized bounding boxes for object tracking of airplane, train, and ship with a bigger size than the vehicle exist in other tracking methods instead of TSTrans. In general, the proposed TSTrans visually demonstrates more robust and accurate tracking performance.

C. Experiment 2 (Ablation Studies)

To validate the effectiveness of individual and paired components including long-term sequential target modeling (*a*), motional correction (*b*) and tracking head pruning (*c*), we implement sufficient ablation experiments on SatSOT [8]: baseline (OSTrack [26] in the first row), TSTrans removing *b* and *c* (the second row), TSTrans removing *a* and *c* (the third row), TSTrans removing *a* and *b* (the fourth row), TSTrans removing *c* (the fifth row), TSTrans removing *b* (the sixth row), TSTrans removing *a* (the seventh row), and TSTrans (ours in the second last row).

As illustrated in Table II, adding an individual component of *a* brings considerable model improvements, proving the validity of our central design. Moreover, paired components of *a* and *c* or *b* and *c* also benefit the tracker, and the combined application of the three components can yield the best performance. In addition, *c*'s removal of the size regression branch is accompanied by fixing the bounding box size with the initially given value, which guarantees proper cropping scale of SR and appropriate model visual range during tracking. Therefore, when adding *b* without *c*, the model's tracking module will tend to suffer from blur or incomplete objects under oversized or too small visual range (without *c*), causing the model's excessive reliance on the motion module (with *b*) and the corresponding partial degradation. This point explains why the model performs slightly poorly in the third and fifth rows compared with the baseline. Also, when adding *c* without *a*, the model is more easily confused by target-similar objects or background components under normal visual range (with *c*), if lack of enough target-aware temporal information (without *a*). This point explains the slight performance decline in the fourth row, contrasted with the baseline.

Specifically, long-term sequential target modeling introduces target changing tendency and robust object understanding through dense relation learning among SR and multiple templates in temporal sequence. Thus, in the second row of Table II, AUC-P is evidently increased by 3.79%, and AUC-S is increased by 3.63%, compared with the baseline in the first row. Subsequently, motional correction plays a role of global refinement for predicted target position through historical trajectory simulation, leading to obvious enhancements of 2.61%/0.79%/1.37% in AUC-P, AUC-S, and Prec-5, comparing the fourth and seventh row. Besides, the trick of tracking head pruning removes the commonly used bounding box regression to put more focus on target localization, resulting in gains of 1.58%/4.4%/1.06% in AUC-P, AUC-S, and Prec-5, comparing the third and seventh row.

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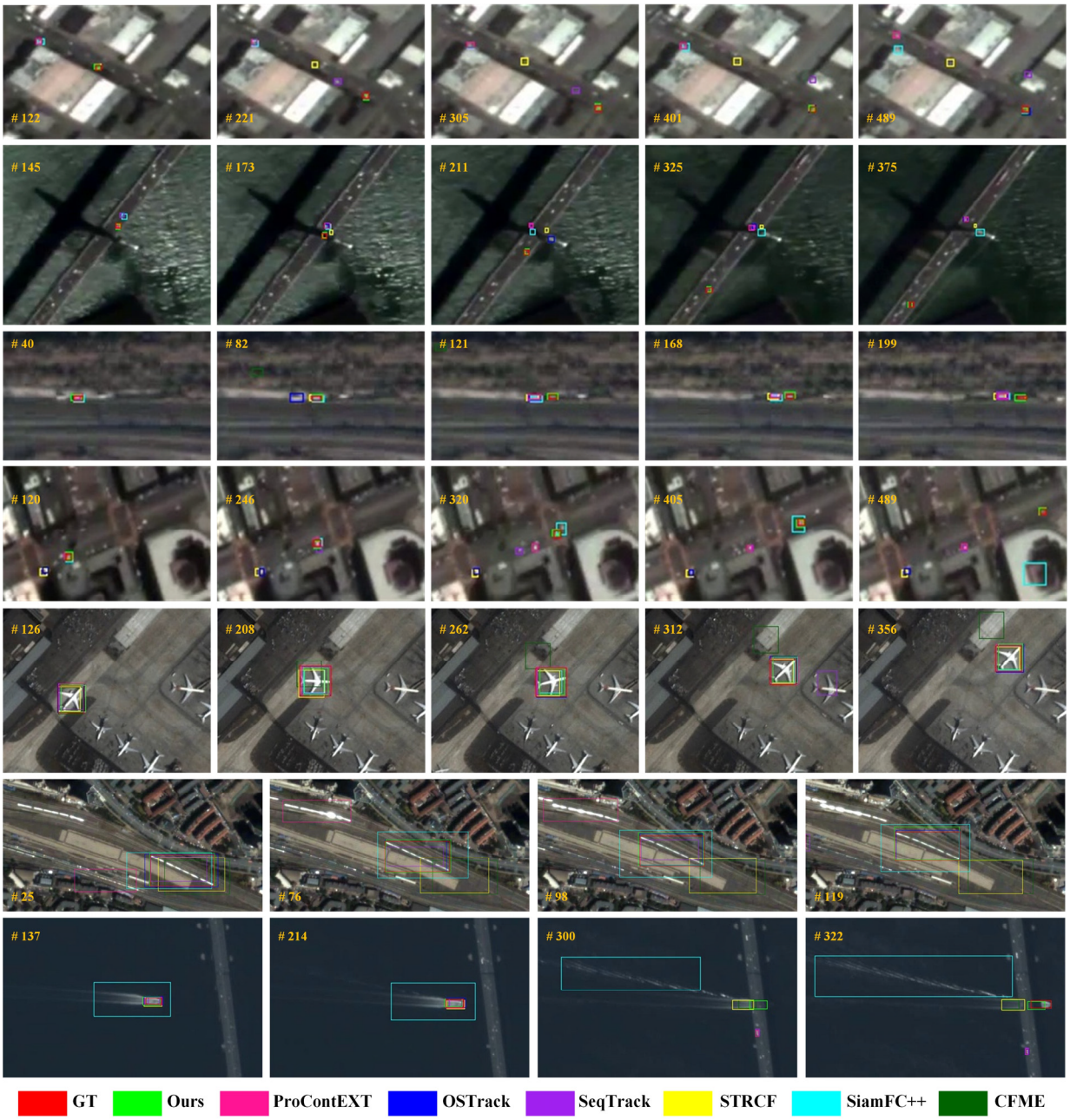


Fig. 7. Visual comparison of our proposed TSTrans with six representative SOTA trackers in seven testing video sequences. The displayed images are cut parts of raw frames. Each row demonstrates frames in different stages from the same video sequence, and their frame IDs are attached to the upper left corner or the bottom left corner.

D. Experiment 3 (Hyperparameters Analysis)

In this section, we present the analysis of these hyperparameters on SV248S [9]: number of temporal templates L and TT confidence threshold α in temporal templates sliding selection, historical reference's length β and attention distraction threshold γ in motional correction module.

For the number of temporal templates controlling the long-term temporal template sequence length and temporal information engagement, we evaluate TSTrans's performance under different values of L in the range of $[1, 5]$ with uniform difference to be 1, and the quantitative results are shown in Table VII. When the number gets lower, temporal templates are too few to span a long range of video frames and learn a complete target variation pattern. When the number gets higher, more temporal templates from historical frames may include some unnecessary ones, which could introduce noise to proper target perception. As listed in Table VII, we set L to be the optimal value 3.

For TT confidence threshold controlling the temporal templates' reliability and updating frequency, we evaluate TSTrans's performance under different values of α in the range of $[0.1, 0.9]$ with uniform difference to be 0.1, and the quantitative results are shown in Table I. When the threshold is too low, frames with inaccurate localizing results will be selected as temporal templates with incomplete or false target patches, introducing more noise. When the threshold is close to 1, there will be hardly qualified frames for selection, causing nearly fixed templates and temporal information. As

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TABLE V

COMPARISON OF TSTRANS WITH 12 ADVANCED TRACKERS OF DIFFERENT CATEGORIES ON SATSOT [8], SV248S [9], VISO [10], AND ALL VIDEO SEQUENCES IN THE THREE TESTING SETS (NAMED "OVERALL"). "CNN-TRANS" DENOTES CNN-TRANSFORMER TRACKER. THE BEST THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE FONTS IN TURN

Categories	Trackers	SatSOT				SV248S			
		AUC-P (%)	AUC-S (%)	Prec-5 (%)	FPS	AUC-P (%)	AUC-S (%)	Prec-5 (%)	FPS
Correlation Filter	KCF	35.64	23.36	20.66	**66.41**	19.94	2.61	4.92	**167.59**
	ECO	62.44	45.76	57.14	5.27	72.74	28.49	61.17	11.36
	STRCF	55.06	41.48	48.66	15.95	**79.50**	**34.99**	**76.65**	16.72
	CFME	62.89	**48.44**	**59.84**	6.19	36.10	22.24	31.72	7.19
Siamese Network	SiamRPN++	41.62	19.20	25.65	35.70	38.53	9.05	22.67	**51.59**
	SiamFC++	57.67	39.15	47.91	**116.48**	74.55	**36.94**	67.10	**142.41**
CNN-Trans	STARK-ST101	46.77	29.60	42.13	17.94	63.28	25.30	48.66	22.99
Fully-Transformer	OSTrack (baseline)	65.87	46.59	57.91	49.61	74.25	32.83	63.96	40.76
	ProContEXT	66.19	**47.12**	**57.95**	45.88	75.76	34.69	68.37	43.22
	SeqTrack	**67.20**	46.82	57.69	31.86	71.90	30.81	59.15	25.24
	ODTrack	39.97	26.46	30.11	46.63	75.12	32.70	66.82	30.69
	LMTrack	**67.32**	46.97	56.89	**51.10**	**79.11**	33.24	**68.95**	51.43
	TSTrans (Ours)	**71.00**	**53.20**	**61.43**	48.52	**86.36**	**40.84**	**80.62**	50.92

Categories	Trackers	VISO				Overall			
		AUC-P (%)	AUC-S (%)	Prec-5 (%)	FPS	AUC-P (%)	AUC-S (%)	Prec-5 (%)	FPS
Correlation Filter	KCF	27.00	4.69	8.48	**151.79**	26.90	5.08	8.63	**128.60**
	ECO	50.09	14.05	33.65	10.85	51.46	15.54	35.53	9.16
	STRCF	**63.42**	**27.32**	**50.91**	16.90	**63.95**	**28.04**	**52.05**	16.52
	CFME	53.66	21.29	38.15	6.79	53.08	22.03	38.41	6.73
Siamese Network	SiamRPN++	35.89	7.66	19.15	41.24	36.16	8.02	19.48	41.57
	SiamFC++	55.84	20.86	41.27	**174.97**	56.75	22.08	42.64	**171.95**
CNN-Trans	STARK-ST101	34.28	13.87	24.62	15.34	35.95	14.81	26.18	15.76
Fully-Transformer	OSTrack (baseline)	49.04	20.12	39.37	23.06	50.64	21.39	40.99	24.56
	ProContEXT	57.45	22.77	45.77	22.86	58.52	23.95	47.13	24.40
	SeqTrack	60.88	**25.03**	46.39	17.77	61.55	25.85	47.28	18.48
	ODTrack	22.18	8.37	16.36	53.15	25.09	9.96	19.05	51.94
	LMTrack	**62.15**	24.95	**50.42**	**55.25**	**63.07**	**25.90**	**51.45**	**54.97**
	TSTrans (Ours)	**65.38**	**28.05**	**51.80**	49.17	**66.50**	**29.29**	**53.39**	49.23

TABLE VI

COMPLEXITY COMPARISON OF TSTRANS WITH EIGHT ADVANCED TRACKERS

Trackers	MACs (G)	Params (MB)	Trackers	MACs (G)	Params (MB)
SiamRPN++	59.604	53.951	SeqTrack	65.862	89.111
SiamFC++	6.898	10.848	ODTrack	32.559	92.121
STRAK-ST101	20.386	47.167	LMTrack	33.134	92.121
OSTrack (baseline)	21.517	92.121	**TSTrans (Ours)**	37.400	89.963
ProContEXT	37.953	92.121			

listed in Table I, we determine α to be the optimal value to be 0.5.

For historical reference's length controlling the number of historical frames employed in motion modeling, we assess TSTrans's performance under different values of β in the range of $[16, 128]$ with uniform difference to be 16, in Table VIII. If the length is too short, the simulated trajectory will be unreliable due to a lack of statistical data. If the length is too long, covering an improper time interval, the target's motion will tend to change greatly, which may impact the historical data's reference value. As listed in Table VIII, we decide β to be the optimal value 64.

For attention distraction threshold controlling whether the model attention is distracted (/whether to correct model prediction) or not, we evaluate TSTrans's performance under different values of γ in the range of $[1, 9]$ with uniform

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TABLE VII
NUMBER OF TEMPORAL TEMPLATES L ADJUSTMENT

Dataset	Methods	AUC-P (%)	AUC-S (%)	Prec-5 (%)
SV248S	$L = 1$	84.45	39.55	77.94
	$L = 2$	83.22	38.92	75.67
	$\boldsymbol{L = 3}$	**86.36**	**40.84**	**80.62**
	$L = 4$	82.71	38.58	75.80
	$L = 5$	85.20	40.39	79.45

TABLE VIII
HISTORICAL REFERENCE'S LENGTH β ADJUSTMENT

Dataset	Methods	AUC-P (%)	AUC-S (%)	Prec-5 (%)
SV248S	$\beta = 16$	**86.17**	**40.71**	**80.82**
	$\beta = 32$	**86.40**	**40.65**	**80.27**
	$\beta = 48$	84.88	39.91	78.91
	$\boldsymbol{\beta = 64}$	**86.36**	**40.84**	**80.62**
	$\beta = 80$	85.06	39.79	78.64
	$\beta = 96$	85.26	40.17	79.18
	$\beta = 112$	85.29	40.17	79.49
	$\beta = 128$	85.20	39.88	78.94

TABLE IX
ATTENTION DISTRACTION THRESHOLD γ ADJUSTMENT

Dataset	Methods	AUC-P (%)	AUC-S (%)	Prec-5 (%)
SV248S	$\gamma = 1$	76.12	23.89	47.28
	$\gamma = 3$	**86.42**	**40.14**	**79.51**
	$\boldsymbol{\gamma = 5}$	**86.36**	**40.84**	**80.62**
	$\gamma = 7$	78.25	34.98	68.57
	$\gamma = 9$	75.10	33.89	66.19

difference to be 2, displayed in Table IX. This threshold influences the dependence degree of motional results, calling for a moderate value. As listed in Table IX, we set γ to be the optimal value 5, which is consistent with the location error threshold of the precision metric for satellite video objects.

E. Discussion

1) Generalization Performance: To evaluate the generalization performance of our algorithm design and hyperparameter setting on unseen datasets, we test a self-annotated dataset, sv20data, using checkpoints trained on the SV248S [9] dataset. Table X demonstrates TSTrans's excellent generalization performance, showing obvious improvement over baseline OSTrack [26] under identical conditions. The sv20data dataset comprises 20 satellite video sequences from Chang Guang and Deimos Imaging, with an average of 378 frames per sequence, ranging from 144 to 750 frames. Objects include eight cars, seven planes, three ships, and two trains, with areas ranging from 16 to 2400 pixels. Some objects in the sv20data are displayed in Fig. 8.

2) Limitations and Future Work: Here we discuss three limitations about this work and how to address them in future research.

TABLE X
GENERALIZATION RESULTS ON SV20DATA

Dataset	Methods	AUC-P (%)	AUC-S (%)	Prec-5 (%)
sv20data	OSTrack	75.80	45.91	67.40
	TSTrans	**91.41**	**66.84**	**91.07**

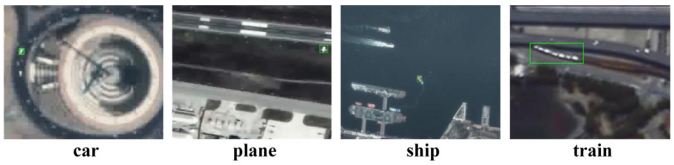


Fig. 8. Some targets of the sv20data dataset in the cropped initial frames. (Labeled in green boxes.)

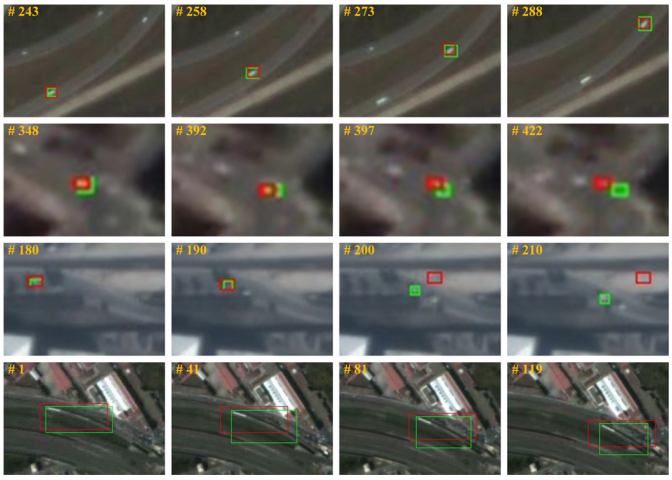


Fig. 9. Visualization for target motions and shape change. (Green box: GT, red box: TSTrans.)

1) Although the motional correction module's assumption of short-term uniformly linear motion [44] has been widely applied, there are limitations of possible tracking failures in specific scenarios, where the target performs nonlinear motion within a short interval, such as sudden acceleration or abrupt turns, rendering the assumption no longer valid. As shown in Fig. 9, we visualize several cases of targets in different motion states. The first row depicts typical target movements that conform to the assumption, where targets in both straight-line and irregular curved paths can be successfully tracked. The second row shows the scenario where the target suddenly accelerates (within 1 s), causing tracking lag. The third row illustrates the case where the target abruptly turns at a large angle (within 1 s), resulting in tracking error. Therefore, in future work, we will explore more adaptive motion models to handle short-term nonlinear movements of the target.
2) Fixed bounding box size is limited in a few scenarios where target shapes undergo significant changes. As shown in the fourth row of Fig. 9, the train's aspect ratio changes when it turns. Applying a fixed bounding box size in such cases will reduce IoU-based metric values.

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Therefore, in future work, we will explore accurate bounding box size regression methods without compromising the overall model optimization.

3) This work employs certain hyperparameters with manual adjustment, which may limit its applicability when transferred to other data domains. Therefore, in future work, we will explore alternative algorithms with more flexibility or adaptive parameter tuning.

V. Conclusion

In this article, we propose a novel temporal-sequence-driven Transformer for SOT in satellite videos.

Driven by target-changing tendency and complete target perception implied in template sequence, we introduce multiple temporal templates to carry dense target-aware attention with SR patches. Driven by motion continuity implied in the historical frame sequence, we predict the target moving trajectory to correct the false tracking position caused by surrounding distractors. Experimental results show that the proposed TSTrans outperforms 12 SOTA trackers, with a running speed of around 50 FPS.

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